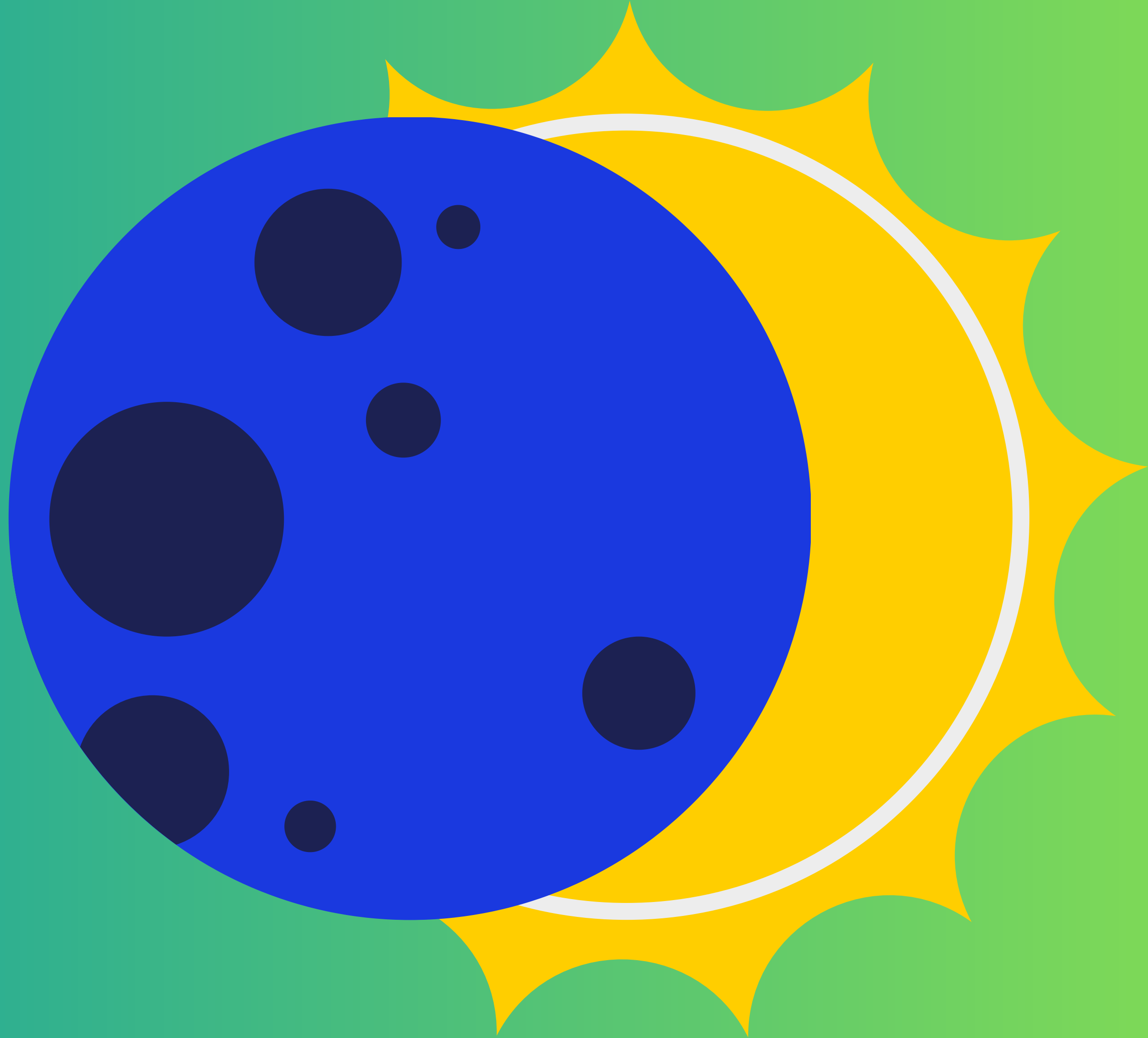
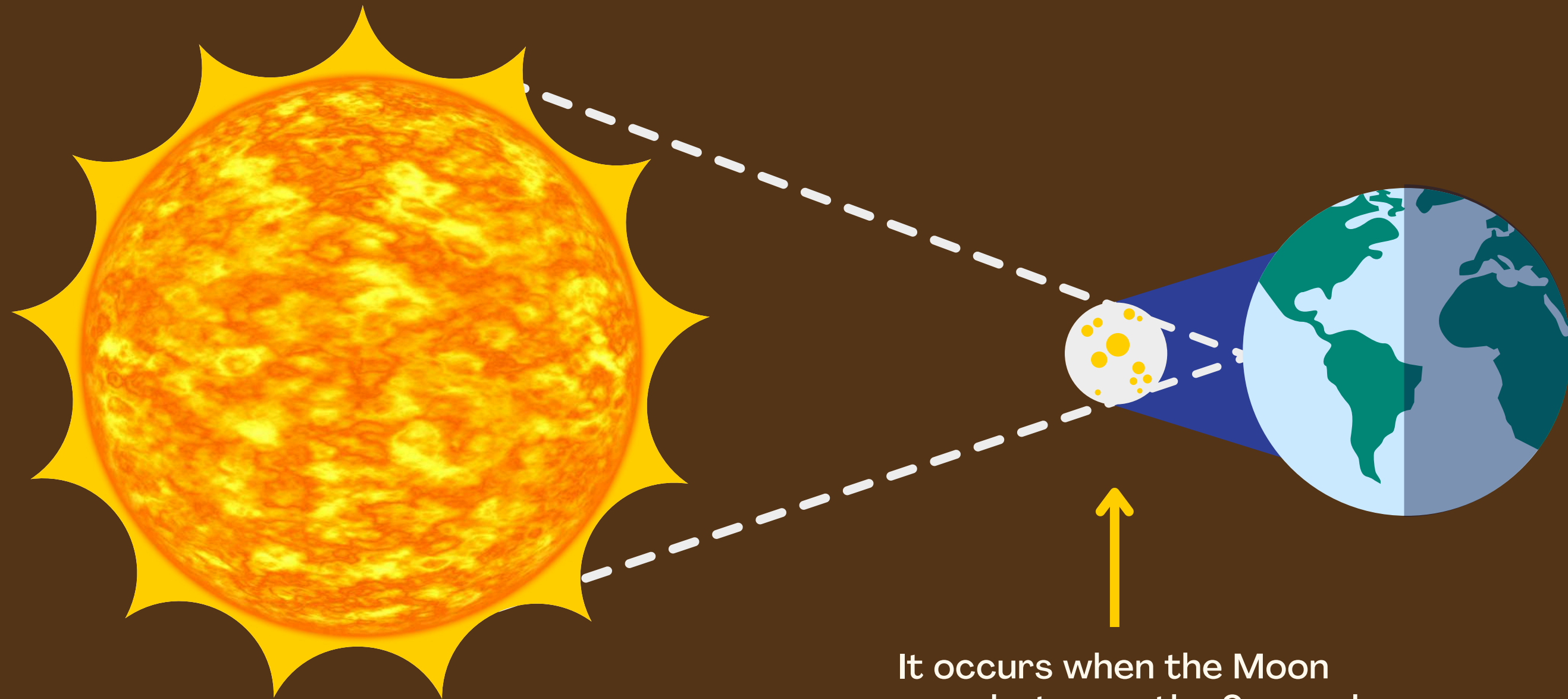


Solar Eclipse

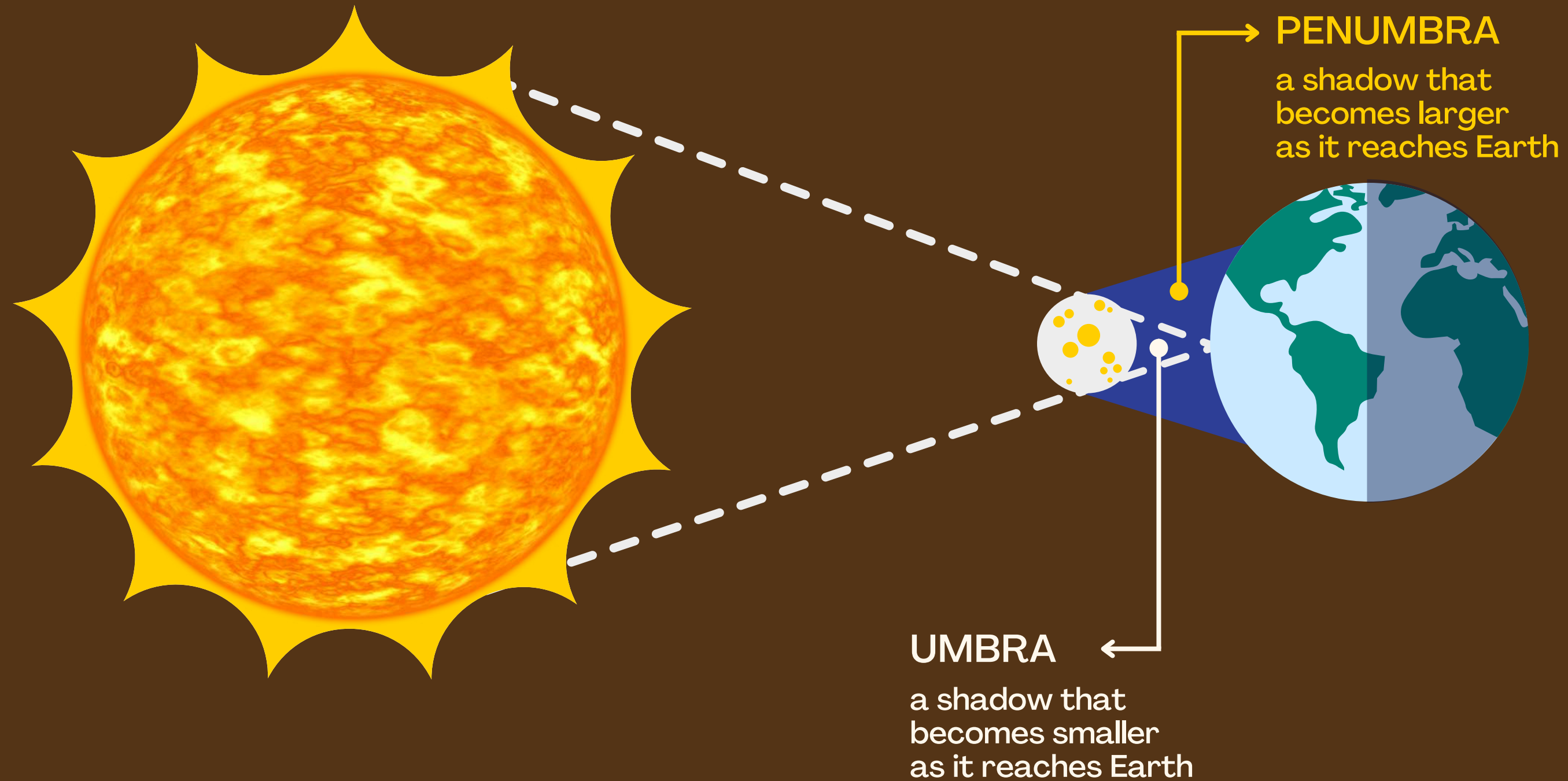


Solar Eclipse



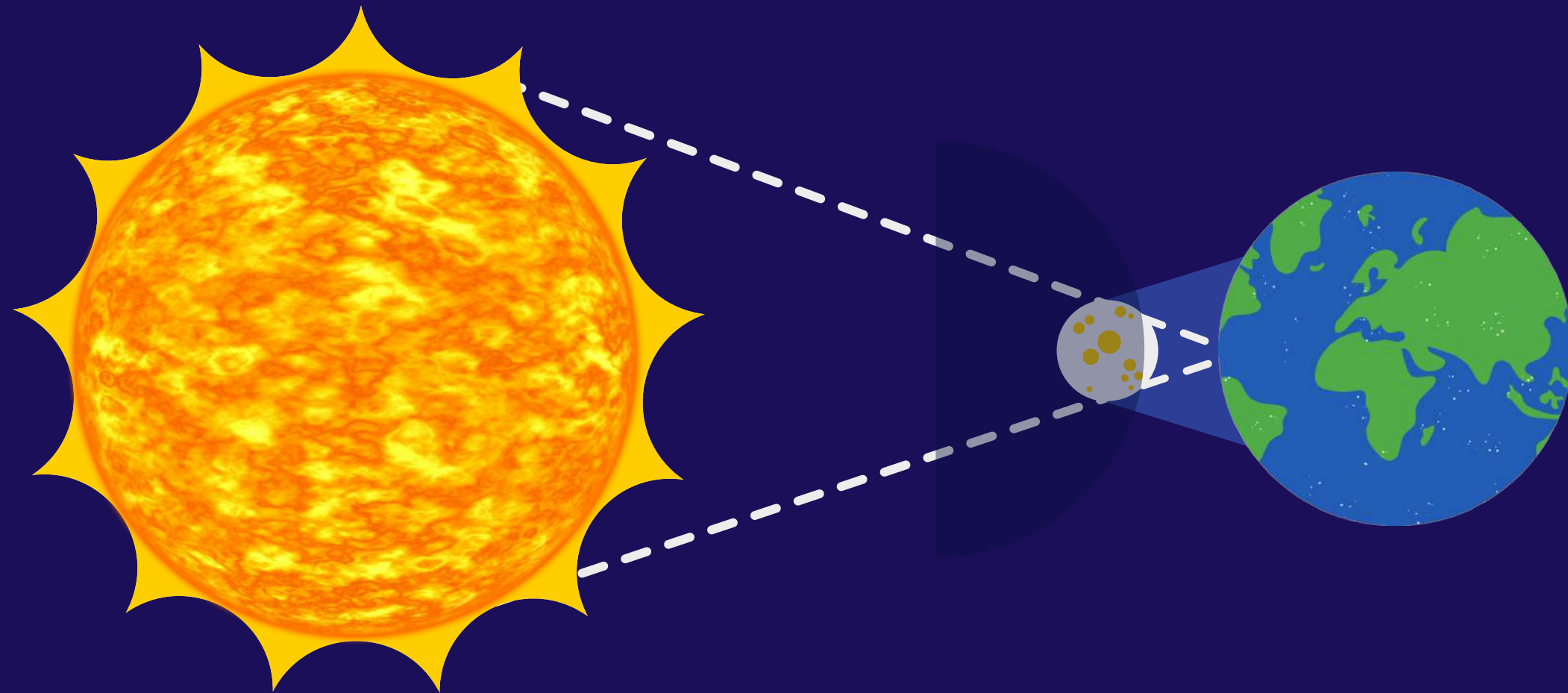
It occurs when the Moon moves between the Sun and Earth. The Moon blocks the light from the Sun and casts a shadow onto Earth.

Solar Eclipse



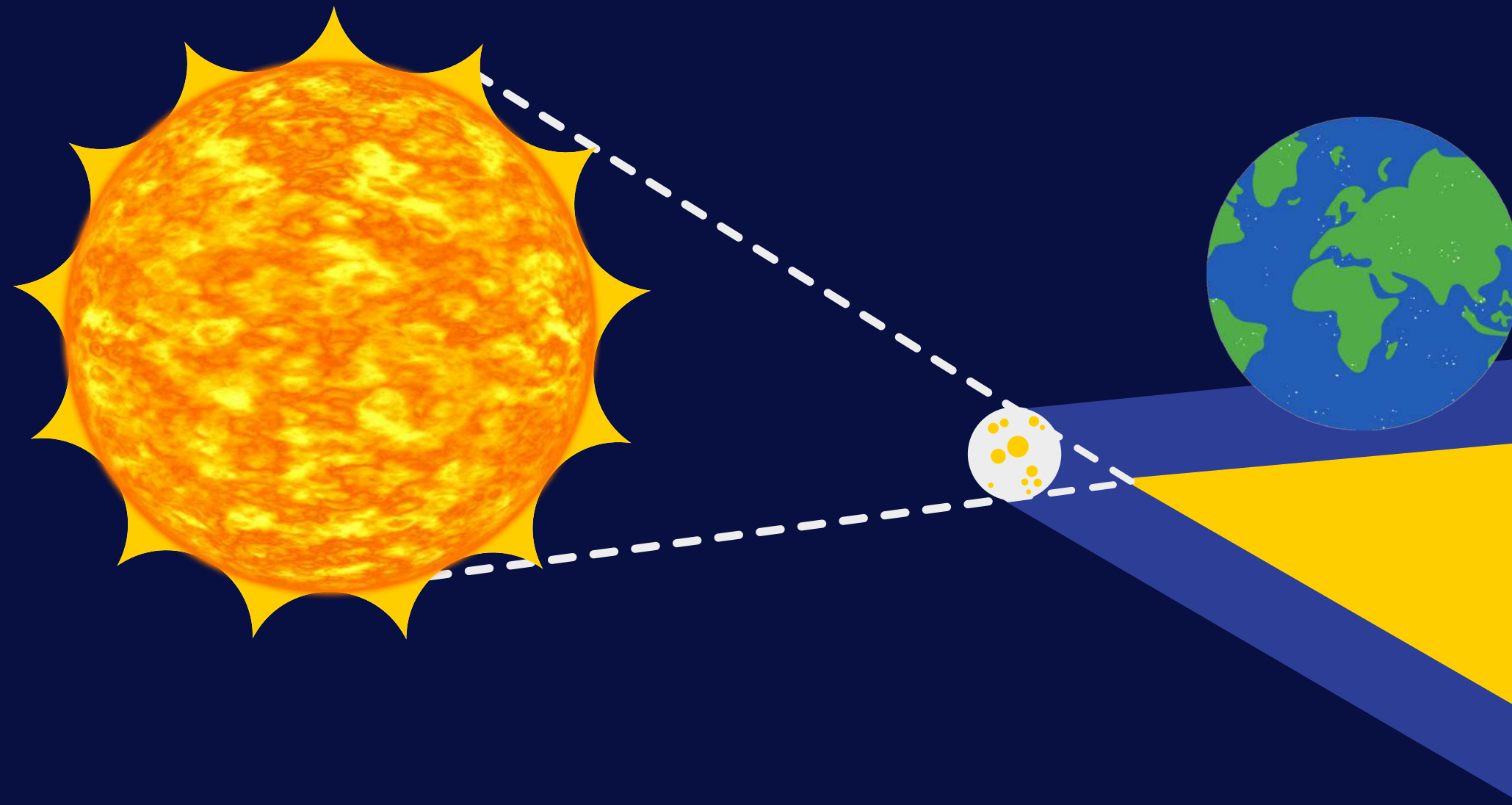
Types of Solar Eclipses

Total Solar Eclipse



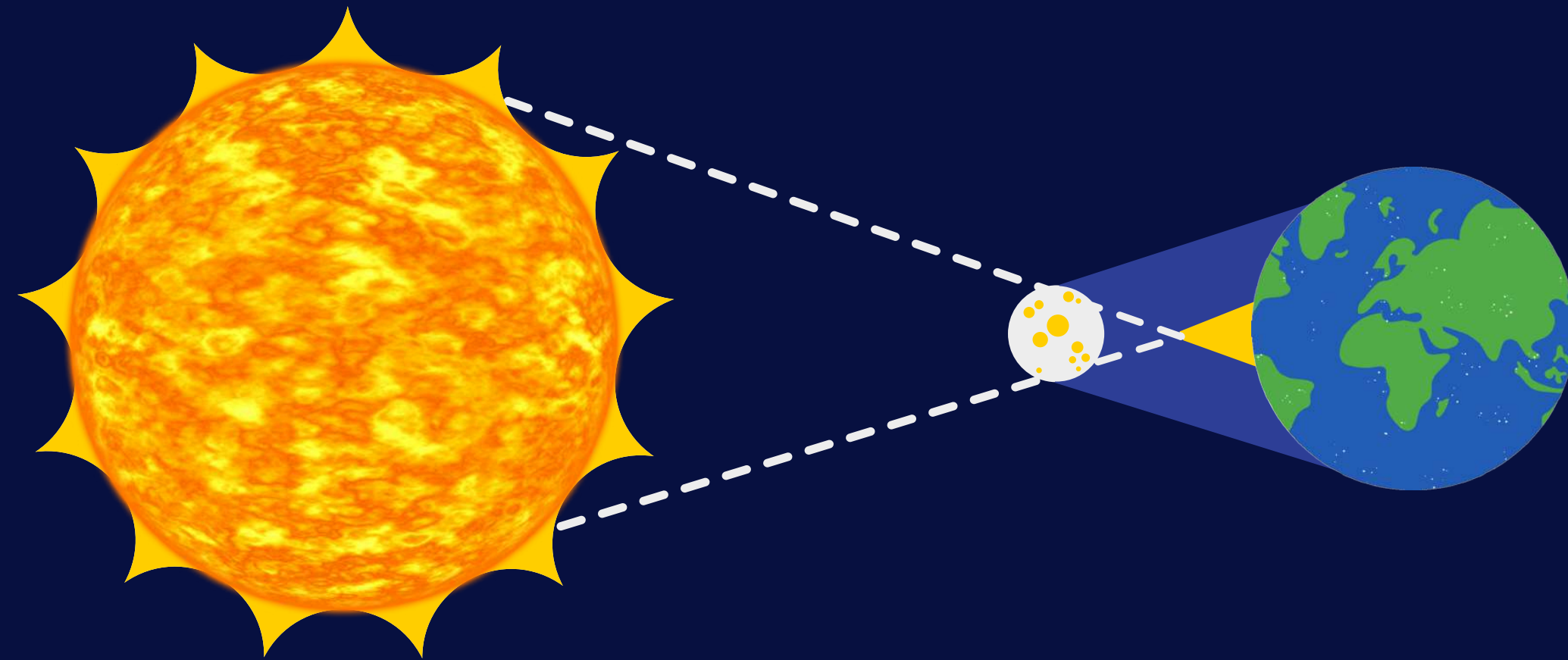
The Sun, Moon, and Earth are aligned perfectly in a line.

Partial Solar Eclipse



The Sun, Moon, and Earth are not exactly aligned.

Annular Solar Eclipse



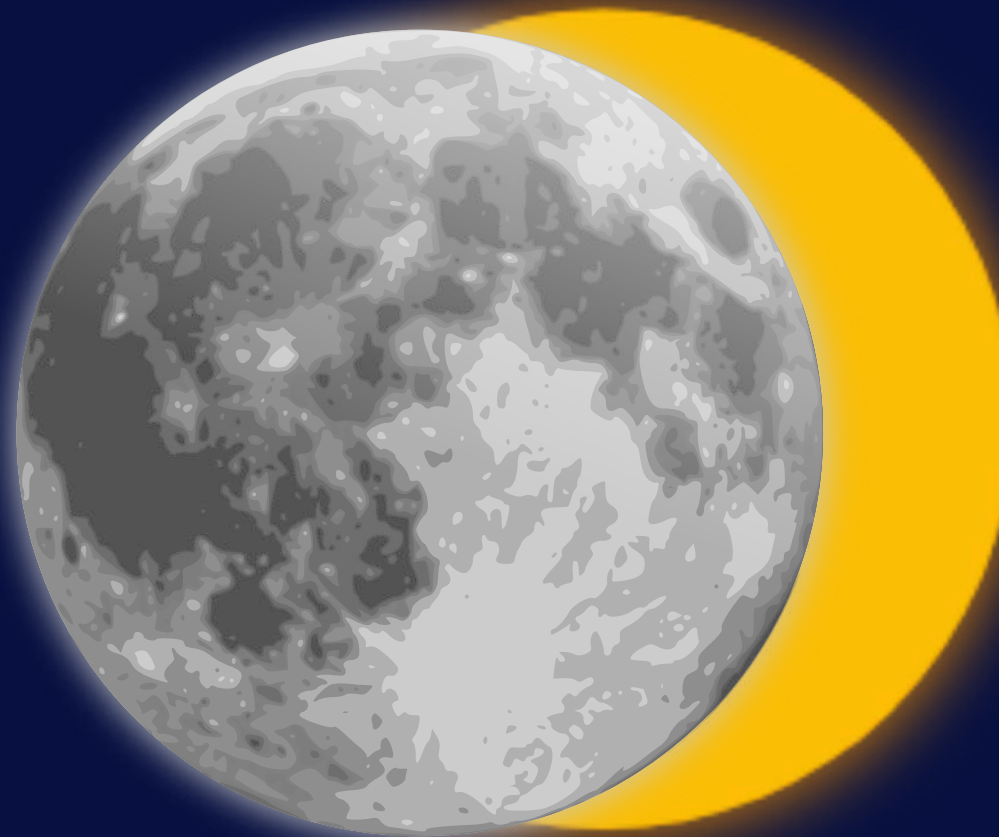
The Sun, Moon, and Earth are not exactly aligned.

Total



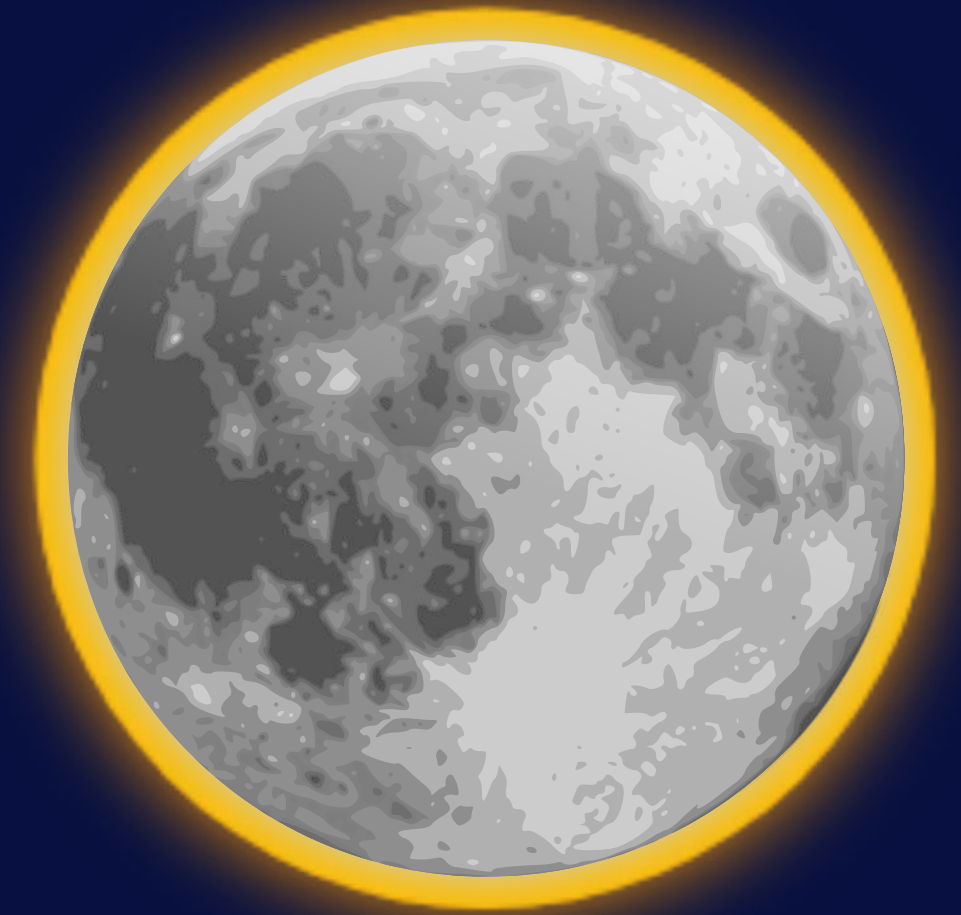
The Moon completely blocks the Sun and Sun's corona becomes visible.

Partial



A small portion of the Sun can still be seen.

Annular



The Moon does not completely block the Sun but creates a ring around the Moon.

Recreate

Creating a solar eclipse simulation at home can be a fun and educational activity for kids. Here's a simple way to do it:

Materials needed:

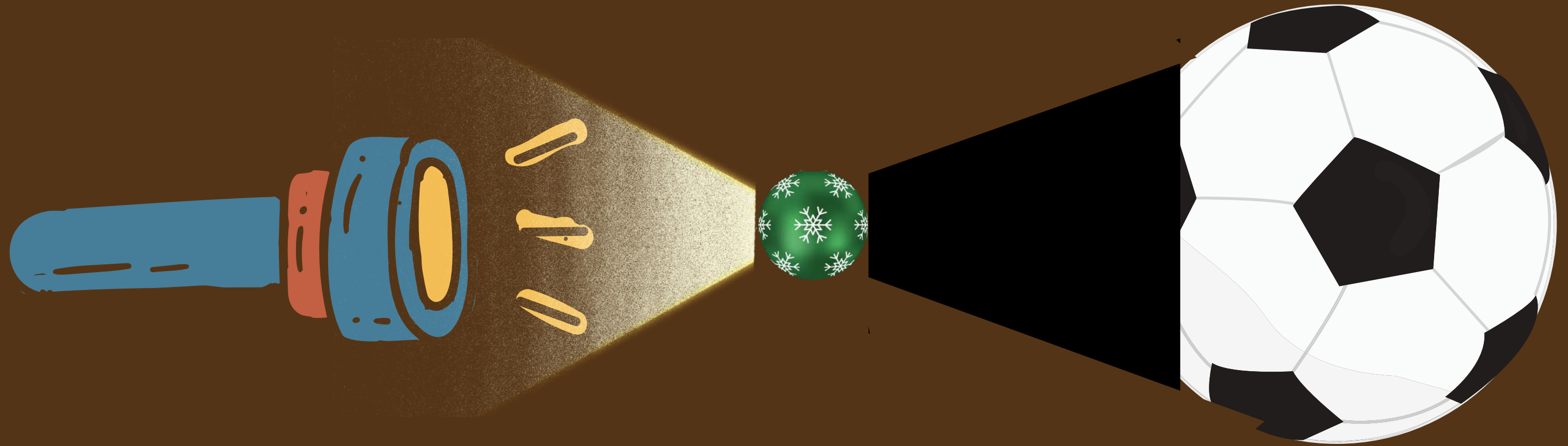
- Flashlight
- Small round object (e.g., a ball or an orange)
- Larger round object (e.g., a plate or a frisbee)
- White wall or ceiling
- Tape

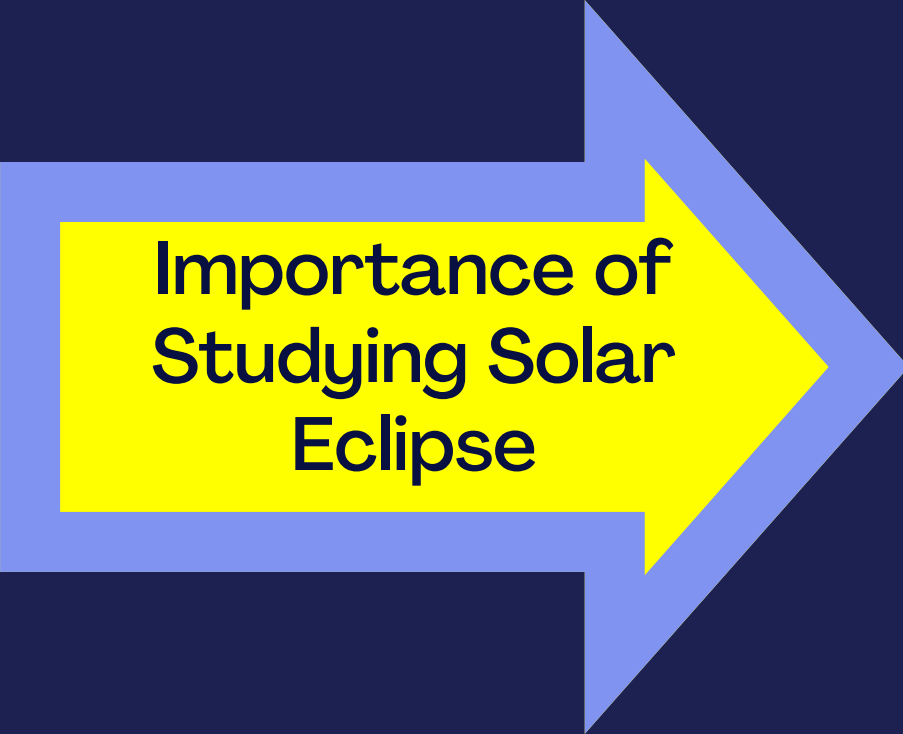
Instructions:

1. Set up your flashlight on a stable surface, aiming it towards a white wall or ceiling in a dark room.
2. Attach the smaller round object (representing the Moon) to the end of a stick or dowel using tape.
3. Position the larger round object (representing the Earth) between the flashlight and the wall/ceiling, ensuring it blocks the direct light from the flashlight.
4. Hold the smaller round object (Moon) on the opposite side of the larger round object (Earth) from the flashlight.
5. Adjust the angle and distance of the Moon from the Earth until its shadow is cast onto the wall/ceiling, creating the appearance of a solar eclipse.
6. Experiment with moving the Moon closer to or farther from the Earth to simulate partial and total solar eclipses.

By adjusting the positioning of the flashlight, smaller round object (Moon), and larger round object (Earth), kids can observe how the alignment of celestial bodies creates different types of solar eclipses. This hands-on activity provides a visual representation of the astronomical phenomenon and helps children understand the science behind solar eclipses in a fun and interactive way.

Recreate





Importance of Studying Solar Eclipse

It helps us learn
more about the
Sun and the
amazing things
that happen in
space

Scientists can
use solar eclipses
to study the
Sun's
atmosphere and
how it affects
our planet

Watching eclipses helps
us learn about the
universe and inspires us
to explore science and
space even more
because it's super fun!

But remember,
it's also
important to
stay safe and
protect our eyes
when watching
eclipses.

Eclipse highlights in 2024 & 2025

April 8, 2024

Total Solar Eclipse

October 2, 2024

Annular Solar Eclipse

March 29, 2025

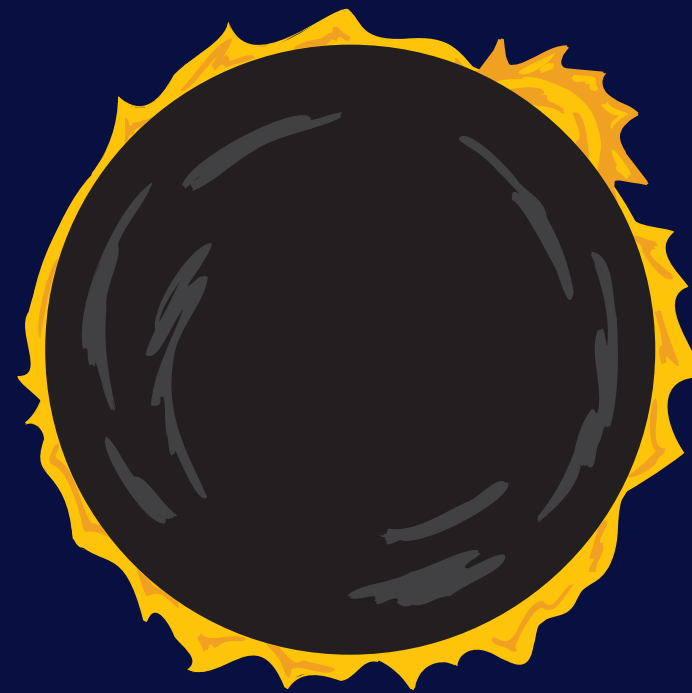
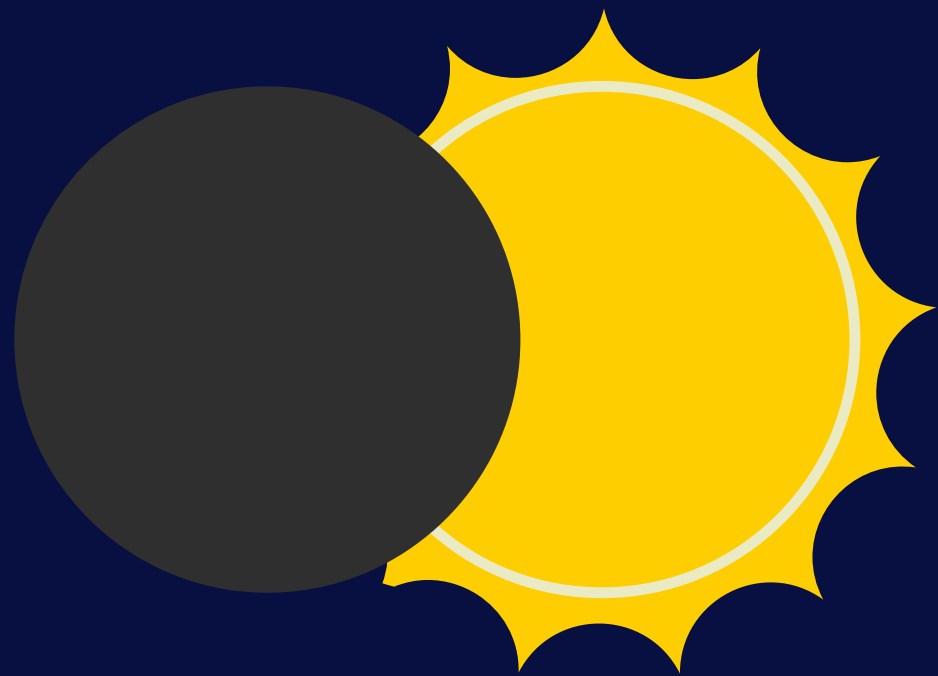
Partial Solar Eclipse

September 21, 2025

Partial Solar Eclipse



Solar Eclipse worksheets



Worksheet 1: Solar Eclipse Vocabulary

Match the term to its correct definition by writing the term number in front of Definition:

1. Solar Eclipse
2. Umbra
3. Penumbra
4. Total Eclipse
5. Partial Eclipse
6. Annular Eclipse

Definitions:

- a. The darkest part of the shadow where the eclipse is total.
- b. A solar eclipse where the Moon covers the Sun's center, leaving a ring of light.
- c. A solar eclipse where only part of the Sun is covered by the Moon.
- d. The lighter part of the shadow where the eclipse is partial.
- e. When the Moon passes between the Earth and the Sun, blocking sunlight.
- f. A solar eclipse where the Sun is completely blocked by the Moon.

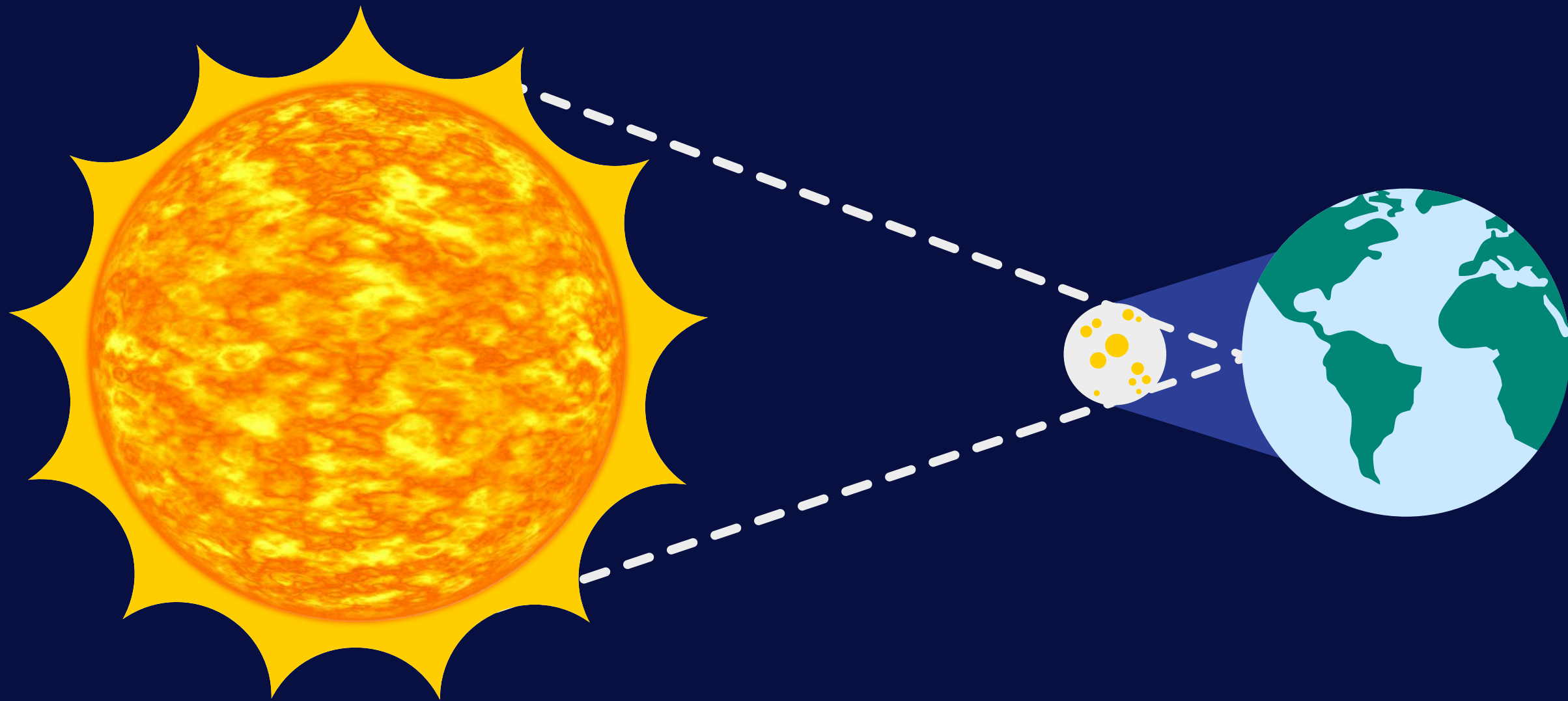
Worksheet 2: True or False

1. A solar eclipse occurs when the Earth moves between the Sun and the Moon.
2. A total solar eclipse can only be seen from the Umbra.
3. Solar eclipses happen every month.
4. You can safely look at a solar eclipse without any protection.
5. An annular eclipse happens when the Moon is farther from Earth in its orbit.
6. The Sun appears completely dark during an annular eclipse.
7. A total solar eclipses always last for several hours.
8. Only people in certain locations on Earth can see a solar eclipse.
9. Animals often behave strangely during a total solar eclipse.
10. You can see the Sun's corona during a total solar eclipse.

Worksheet 3: Label the Diagram

Task:

- Label the positions of the Sun, Moon, and Earth.
- Mark the Umbra and Penumbra.
- Write which type of eclipse occurs in each region (Total or Partial).



Worksheet 4: Fill in the Blanks

1. A solar eclipse happens when the _____ passes between the Earth and the _____.
2. The darkest part of the shadow is called the _____.
3. You should always wear _____ glasses to view a solar eclipse.
4. During a(n) _____ eclipse, the Moon appears smaller than the Sun, leaving a “ring of fire.”
5. Solar eclipses only happen during the _____ phase of the Moon.
6. The lighter part of the Moon’s shadow is called the _____.
7. A solar eclipse is rare because the Moon’s orbit is _____ compared to Earth’s orbit around the Sun.
8. The Sun is completely blocked during a _____ solar eclipse.
9. Ancient civilizations often associated solar eclipses with _____.
10. Special tools like a _____ projector can be used to view a solar eclipse safely.

Worksheet 5: Multiple Choice

1. What phase of the Moon is needed for a solar eclipse?

- a) Full Moon
- b) New Moon
- c) Half Moon
- d) Crescent Moon

2. Which part of the shadow causes a partial eclipse?

- a) Umbra
- b) Penumbra
- c) Solar Flares
- d) Corona

3. How long does a total solar eclipse typically last?

- a) A few seconds
- b) A few minutes
- c) An hour
- d) A day

Worksheet 5: Multiple Choice

4. What is the main reason solar eclipses are rare?
 - a) The Sun moves too quickly
 - b) The Moon's orbit is tilted compared to Earth's orbit
 - c) The Moon is too far away
 - d) Solar eclipses only happen during a Full Moon

5. What phase of the Moon is required for a solar eclipse to occur?
 - a) Full Moon
 - b) New Moon
 - c) Crescent Moon
 - d) Half Moon

6. Which type of solar eclipse allows you to see the "ring of fire"?
 - a) Total eclipse
 - b) Partial eclipse
 - c) Annular eclipse
 - d) Lunar eclipse

Answer Key

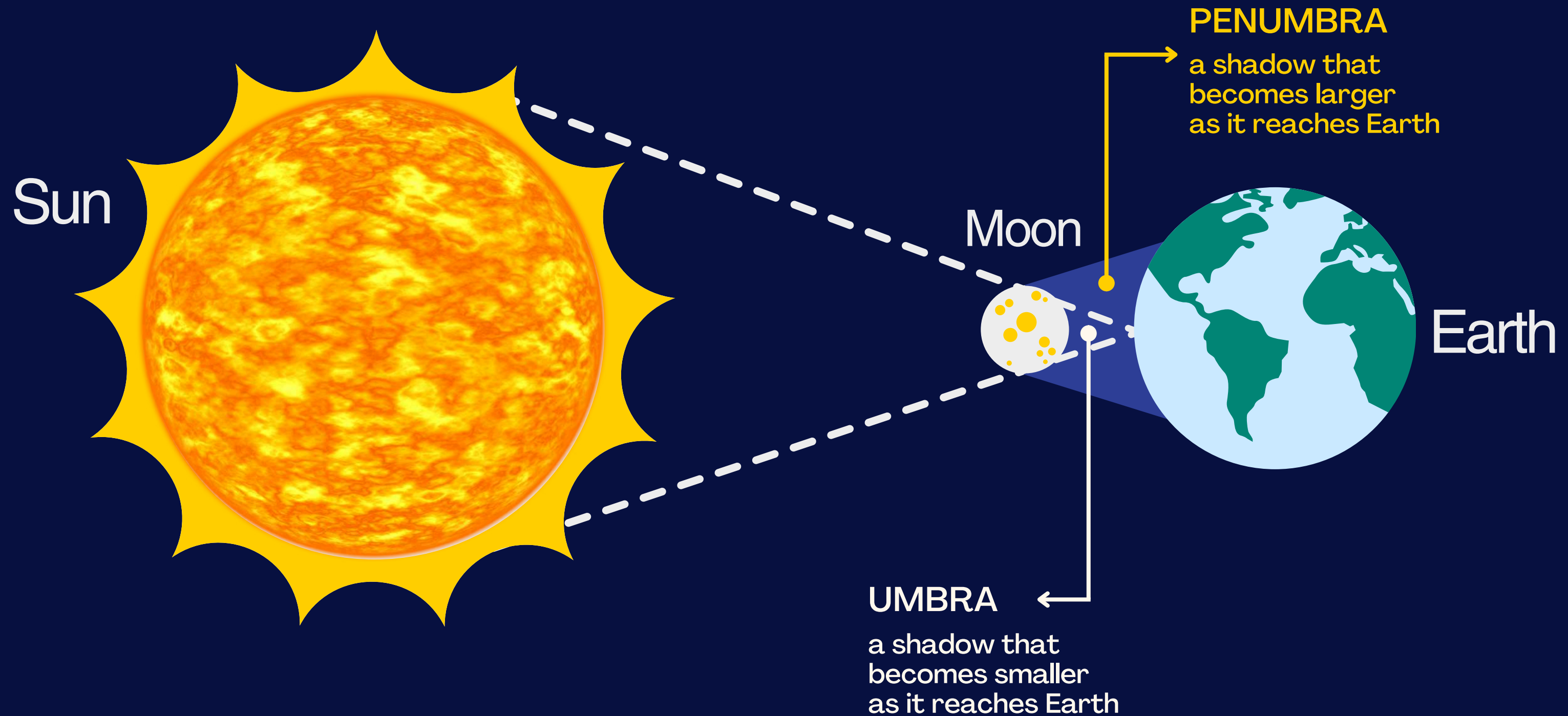
Worksheet 1 - Vocabulary

1. Solar Eclipse - e. When the Moon passes between the Earth and the Sun, blocking sunlight.
2. Umbra - a. The darkest part of the shadow where the eclipse is total.
3. Penumbra - d. The lighter part of the shadow where the eclipse is partial.
4. Total Eclipse - f. A solar eclipse where the Sun is completely blocked by the Moon.
5. Partial Eclipse - c. A solar eclipse where only part of the Sun is covered by the Moon.
6. Annular Eclipse - b. A solar eclipse where the Moon covers the Sun's center, leaving a ring of light.

Worksheet 2 - True or False

1. False – A solar eclipse occurs when the Moon moves between the Sun and the Earth.
2. True – A total solar eclipse can only be seen from the Umbra.
3. False – Solar eclipses don't happen every month because the Moon's orbit is tilted relative to Earth's orbit around the Sun.
4. False – You need special solar viewing glasses to safely look at a solar eclipse.
5. True – An annular eclipse occurs when the Moon is farther from Earth in its orbit, making it appear smaller.
6. False – The Sun is not completely dark during an annular eclipse; a "ring of fire" is visible.
7. False – A total solar eclipse typically lasts only a few minutes in any one location.
8. True – Only certain locations within the Moon's shadow path can see a solar eclipse.
9. True – Animals often exhibit unusual behaviors, such as becoming quiet or preparing for nighttime, during a total solar eclipse.
10. True – The Sun's corona (outer atmosphere) becomes visible during a total solar eclipse.

Answer Key : Worksheet 3: Label the Diagram



It is a Total Solar Eclipse

Answer Key

Worksheet 4: Fill in the Blanks

1. A solar eclipse happens when the Moon passes between the Earth and the Sun.
2. The darkest part of the shadow is called the Umbra.
3. You should always wear solar viewing glasses to view a solar eclipse.
4. During a(n) annular eclipse, the Moon appears smaller than the Sun, leaving a “ring of fire.”
5. Solar eclipses only happen during the New Moon phase of the Moon.
6. The lighter part of the Moon’s shadow is called the Penumbra.
7. A solar eclipse is a rare event because the Moon’s orbit is tilted compared to Earth’s orbit around the Sun.
8. The Sun is completely blocked during a total solar eclipse.
9. Ancient civilizations often associated solar eclipses with supernatural events.
10. Special tools like a pinhole projector can be used to view a solar eclipse safely.

Worksheet 6 - Multiple Choice

1. b) New Moon
2. b) Penumbra
3. b) A few minutes
4. b) The Moon’s orbit is tilted compared to Earth’s orbit
5. b) New Moon
6. c) Annular eclipse